Answers

VdL

MECH 325 - Midterm Book 1

Tuesday, October 21, 2014

Name: _____

Student Number: _____

Circle your section

Section 101:	**T1A (11:00am tutorial)**	**T1B (3:30pm tutorial)**
Section 102:	**T1C (2:00pm tutorial)**	**T1D (5:00pm tutorial)**

Signature: _____

Part 1 MC Mark _____ / 14

Part 1 SA Mark _____ / 16

Part 2 LA Mark _____ / 30

Total _____ / 60

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (30 marks)

Multiple choice (14 marks: 1 marks per question). Complete all 14 questions by marking your response **in pencil in the computer score card**. Write your name and student number on the computer card and mark your student number in the "ID Field".

Short answer (16 marks: 2 or 4 marks per question). Complete all 6 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your booklet and Scantron scorecard and you will receive a handout for Part 2 and a separate exam booklet for your answers. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (30 marks)

Long answer (30 marks): Complete all parts to the questions for Part 2 by marking your responses in the separate exam booklet. This portion of the exam is open-book and open-notes.

Only non-programmable calculators may be used (or programmable calculators with purged memories).

Part 1A: Multiple Choice

Answer all 14 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

1. **Which of the following types of gears is appropriate for high-efficiency, smooth (low-noise) transmission of motion between parallel shafts with minimal axial loading on either shaft?**
   a) Bevel
   b) Spur
   c) Worm
   ✓ d) Herringbone
   e) Helical

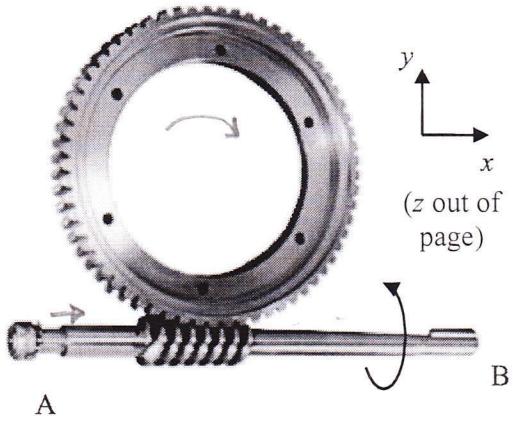
2. **A spur gear has 40 teeth, a 14° pressure angle, and a pitch diameter of 2.0 in. What is the diametral pitch of the gear?**
   a) 0.05
   b) 1.42
   c) 2.86
   d) 7.0
   ✓ e) 20.0

$P = \frac{N}{d} = \frac{40}{2} = 20$

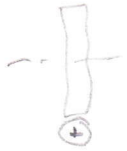
3. **The worm gear shown has a 1" pitch diameter and has two threads. The gear has an 8" pitch diameter and 96 teeth. If the worm turns 24 revolutions in the direction indicated, how far does the gear turn?**
   a) 4 rotations clockwise
   b) 1 rotation clockwise
   ✓ c) ½ rotation clockwise
   d) 1 rotation counter-clockwise
   e) 4 rotations counter-clockwise

$N_g = N_w \cdot \frac{\text{threads}}{\text{teeth}}$

$= 24 \cdot \frac{2}{96} = \frac{1}{2}$



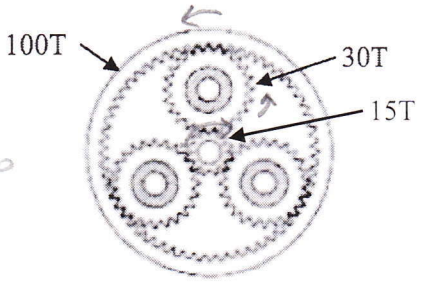
4. **In the worm from Question 3, the bearing at A is the locating bearing. What direction are the forces <u>from</u> the worm shaft <u>onto</u> bearing A?**
   a) - x + y - z
   ✓ b) + x - y + z
   c) + x - y - z
   d) - x - y + z
   e) - x - y - z



5. **A spur gearset has an input gear with 15 teeth and an output gear with 60 teeth. If 4 N·m of torque is supplied to the input gear at 50 rad/sec, approximately how much power is delivered by the output gear to the load?**
   a) 16 W
   b) 60 W
   c) 75 W
   ✓ d) 200 W
   e) 800 W

$H = T \cdot \omega = 4 \cdot 50 = 200\ W$

6. **For the planetary gear train shown, the sun gear rotates at 2000 rpm clockwise and the arm is fixed. How fast does the ring gear rotate and in which direction?**



a) 4,000 rev/min (clockwise)
b) 2,500 rev/min (clockwise)
✓ c) 300 rev/min (counter-clockwise)
d) 300 rev/min (clockwise)
e) 2,500 rev/min (counter-clockwise)

$2000 \cdot \frac{15}{100} = 300$

7. **Comparing a square vs. an ACME power screw, which of the following is true regarding the use of a split-nut:**
a) A split nut reduces friction on both types of power screw
✓ b) A split nut allows for backlash adjustment on an ACME power screw
c) A split nut allows for backlash adjustment on a square power screw
d) A split nut is required when using a bushing for a friction collar on both types of power screw
e) A split nut is required when an ACME power screw is non-reversing

8. **Which of the following components of belt tension present in V-Belts but normally ignored in flat belts that is the main consideration when calculating belt life?**
a) Initial tension (F_i)
✓ b) Bending tension (F_b)
c) Centrifugal tension (F_c)
d) Tight-side tension (F_1)
e) Slack-side tension (F_2)

9. **A V-belt drive has a driving pulley operating at 200 rpm, a drivetrain value of e = 0.25, a nominal power of H_{nom} = .75 hp, a service factor of K_s = 2.5, and a design factor of n_d = 1.5. TWO belts are to be used in parallel. The following belts are available in the catalogue. Which is the smallest belt that will satisfy the design requirements?**
a) Allowable power H_a = 0.7 hp
b) Allowable power H_a = 1.1 hp
✓ c) Allowable power H_a = 1.5 hp
d) Allowable power H_a = 2.9 hp
e) Allowable power H_a = 5.7 hp

$H_a = H_{nom} \cdot K_s n_d / N_b = .75 \cdot \frac{2.5 \cdot 1.5}{2} = 1.4\,HP$

10. **A timing belt has a driving pulley with pitch diameter of 20 cm and 50 teeth, and a driven pulley with 100 teeth. The wrap angle is 130°. During operation, the tight side tension is 130 N, slack side tension is 40 N, and the driving pulley rotation rate is 120 rad/sec. What is the torque transmitted from the motor shaft to the driving pulley?**
a) 6.0 N·m
✓ b) 9.0 N·m
c) 60.0 N·m
d) 100.0 N·m
e) 600 N·m

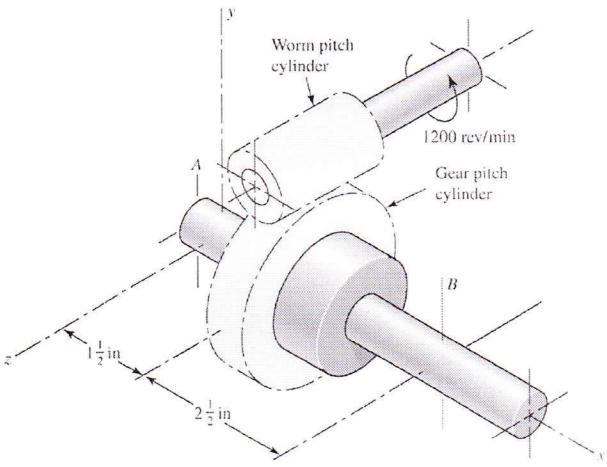
$T = (F_1 - F_2) \cdot d/2 = (130 - 40) \cdot .2/2 = 90\,Nm$

11. **Which of the following design or operating parameters should be increased in order to be able to reduce the required bushing length of a boundary-lubricated bearing in service? (assume all other parameters remain fixed)**
    a) Operating rotation speed, N
    b) Applied force, F
    c) Coefficient of friction, f_s
    d) (PV) rating of the bushing
    e) Wear factor, K

12. **A tapered-roller bearing is best suited for:**
    a) Only large radial loads
    b) Only large axial loads
    c) Small radial loads, plus large axial loads in both directions
    d) Large radial loads, plus large axial loads in both directions
    e) Large radial loads, plus large axial loads in one direction only

13. **Which of the following statements is true for the Soderberg, Gerber, Modified Goodman and ASME-elliptic fatigue failure criteria:**
    a) All criteria predict fatigue failure will occur before yielding
    b) Only the Modified Goodman criterion is valid for tensile stresses
    c) The zone to the left of each of the fatigue criteria lines represents the design domain where $n < 1$
    d) When $\sigma_a = 0$, the safety factor $n \geq 1$ when $\sigma_m \leq S_y$ for all criteria
    e) When $\sigma_m = 0$, the safety factor $n = 1$ when $\sigma_a = 0.5 * S_e$ for all criteria

14. **A motor drives a gearbox input shaft (top shaft in figure to right) at 1200 rpm. The gearbox uses a worm gear to drive a constant torque load attached to the output shaft via a flexible coupling. In addition to torsional shear and direct shear due to the worm gearset, the output shaft is subject to stresses best characterized by:**
    a) steady bending stress plus alternating axial stress
    b) steady bending stress plus steady axial stress
    c) zero midrange stress and steady axial stress
    d) partially but not fully-reversed bending stress only
    e) alternating bending stress in the z-axis only, plus alternating axial stress

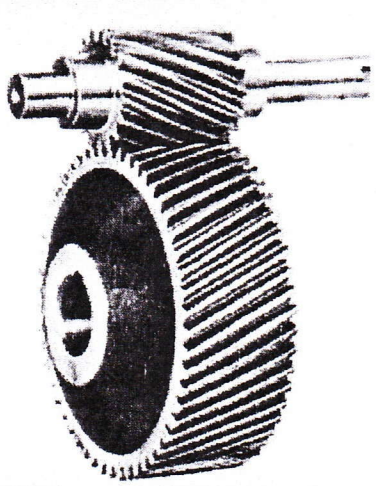


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Part 1B – Short-Answer Questions (16 marks)

Answer all 7 questions in the spaces provided. Each question is worth 2 - 4 marks.

15. The larger helical gear shown has a normal pressure angle of 20°, a helix angle ψ of 30°, N = 35 teeth, and a pitch diameter of 5". What is the normal pitch in inches? Show your calculations. (2 marks)

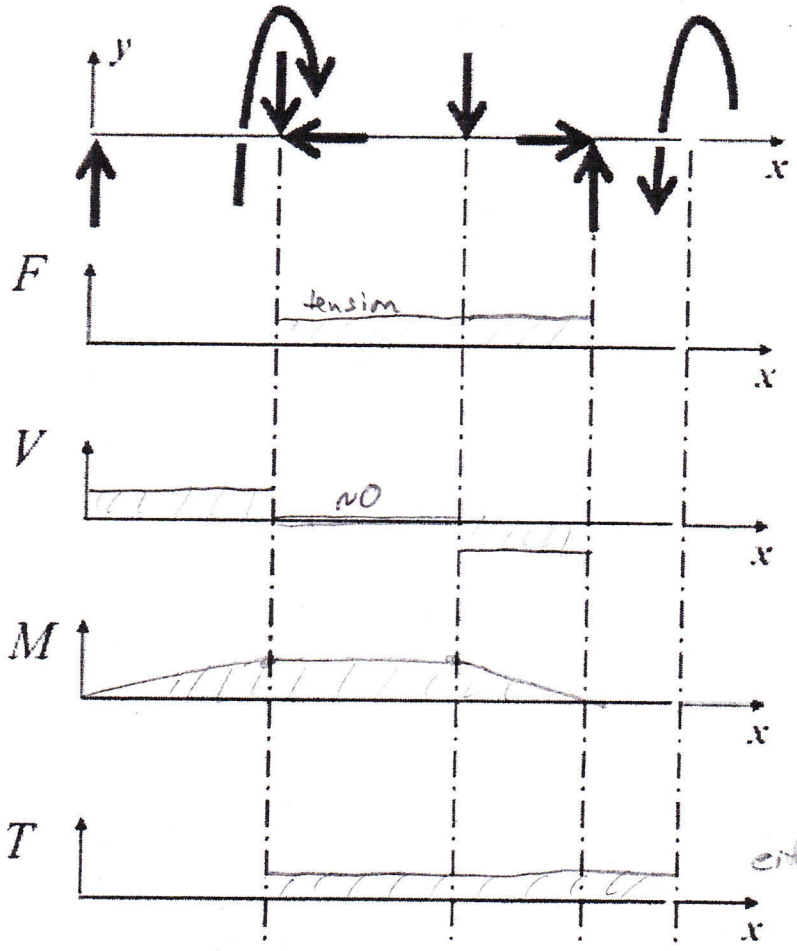


$P_T = \frac{N}{d}$; $P_t P_t = \pi$; $P_n = P_t \cos\psi$

$P_T = \frac{35}{5} = 7$, $P_t = \frac{\pi}{7}$; $P_n = \frac{\pi}{7}\cos 30°$

Normal pitch = 0.388 "

16. Based on the shaft loading shown in the top diagram, complete the diagrams for axial force, shear force, bending moment and torsion. Notes: The moments shown are oriented along the x-axis. You do not need to indicate values in your diagrams, but they must follow the usual sign convention. (4 marks)



① ① ① ①

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17. Given that the total raising torque, T_T, for an ACME power screw is given by the sum of the raising torque at the threads (T_R) and collar friction torque (T_C):

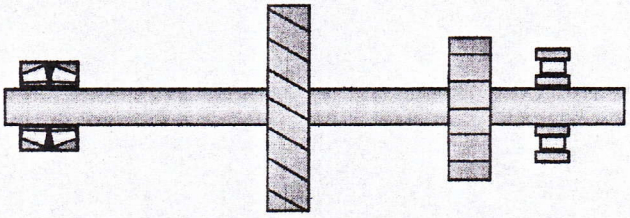
$$T_T = T_R + T_C = \frac{Fd_m}{2}\left(\frac{\pi f d_m \sec\alpha + l}{\pi d_m - f l \sec\alpha}\right) + \frac{Ff_c d_c}{2}$$

What is the efficiency, e, which is defined as the ratio of the total raising torque when friction has been eliminated divided by the total raising torque? Simplify the expression, and leave the denominator as T_T. (2 marks)

$$e = \frac{\frac{Fdm}{2}\left(\frac{l}{\pi dm}\right)}{T_T} = \frac{\frac{Fl}{2\pi}}{T_T} = \frac{Fl}{2\pi T_T}$$

(all ok)

18. The shaft shown has a pair of tapered roller bearings, a helical gear, a spur gear, and a cylindrical roller bearing. Power is transmitted to the large helical gear via a smaller pinion gear and runs in one direction only, with smooth torque delivery. Assuming proper shaft accessories, what are the TWO most significant changes you would suggest to this design and why? (2 marks)



Change #1.

- shorten shaft

or put components closer to bearings

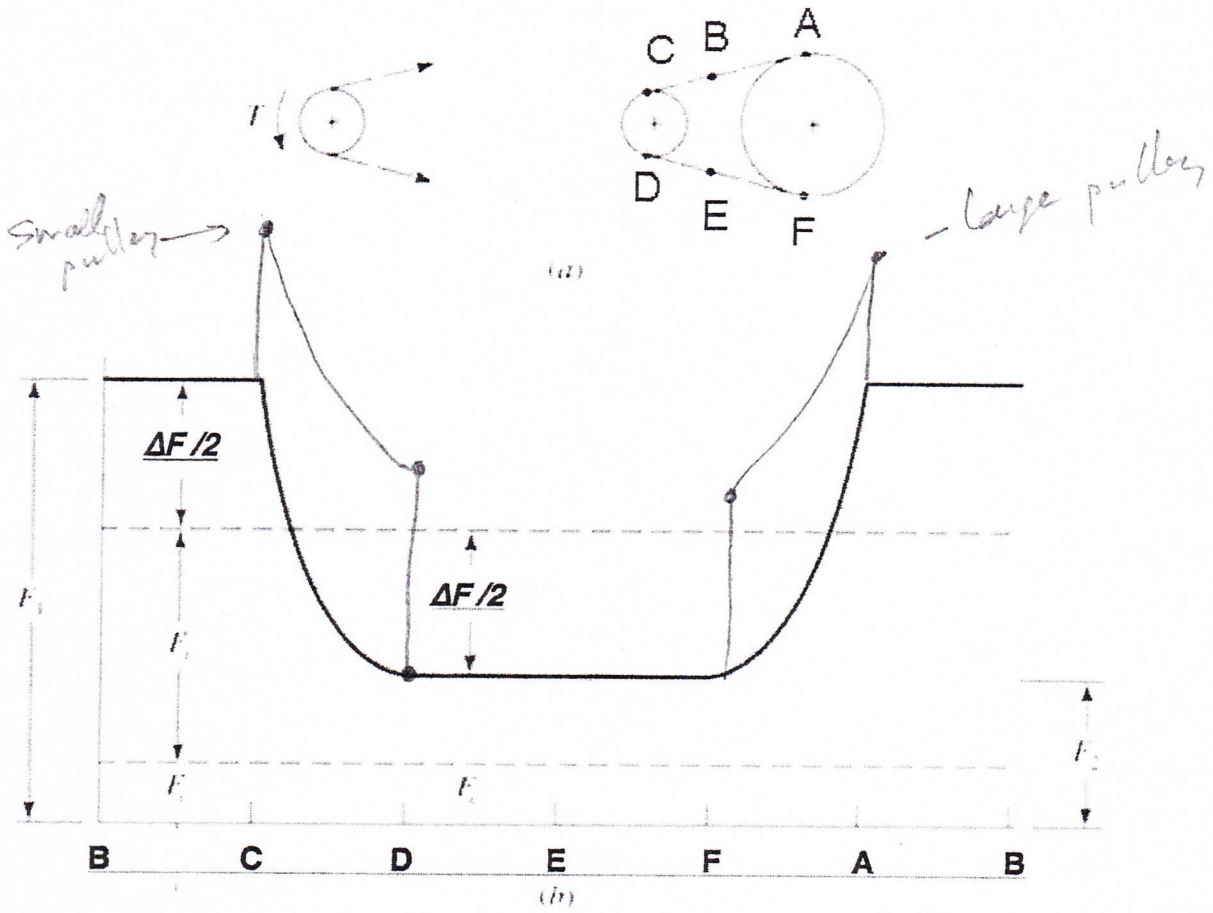
Change #2.

- put components closer to bearings

[ok: add axial constraints, replace tapered roller bearing with angular contact ball bearing.

V dl

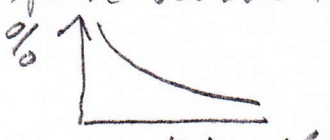
19. The diagram shown below is the tension profile for a flat belt. On this diagram, sketch the tension profile for a V-belt belt that is transmitting the same torque, but otherwise has the same linear speed, initial tension, pulley diameters, and belt weight per unit length. (2 marks)



20. A) For synchronous power transmission with a flexible element, what is the principal *operating* difference between roller chains and timing belts? Exclude materials and secondary effects such as lubrication, dirt, and maximum capacity. (1 mark)

B) What is the simplest way to minimize this difference? (1 mark)

A: roller chains have chordal speed variation



B:
- increase #teeth
- increase diameter of sprocket

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Adil

21. (4)

$H = 2\,KW$

$n = 3000$ RPM $\Rightarrow \omega = 3000 \cdot \frac{2\pi}{60} = 314$ rad/sec

$e = 97\%$

$H_{shaft\ b} = 0.97 \cdot 2000 = 1940\ W$ (1)

$\omega_{shaft\ b} = \frac{2\pi}{60} \cdot n \cdot \frac{N_2}{N_3} = \frac{2\pi}{60} \cdot 3000 \cdot \frac{15}{30} = 157\ r/s$ (1)

$T = \frac{H}{\omega} = \frac{1940}{157} = 12.34\ Nm$ (1)

$V = n \cdot \pi d = 3000 \cdot \frac{15}{30} \cdot \pi \cdot \frac{5}{12} = 1963\ \frac{ft}{min}$ (1)

$V = 1963\ \frac{ft}{min} \cdot .305\ \frac{m}{ft.} \cdot \frac{min}{60\ sec} = 9.98\ \frac{m}{s}$ (OK)

VM

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22. (4)

$T_b = 4.87$ Nm

$\omega = 157$ r/s ; $n = 1500$ RPM

$K_s = 1.3$

$n_d = 1.5$

$H = T\omega = 4.87 \cdot 157 = 0.764$ KW

$H = 0.764 \text{ KW} \cdot \frac{\text{HP}}{0.746 \text{ KW}} = \underline{1.024 \text{ HP}}$ (1)

$H_d = K_s \cdot n_d \cdot H = 1.5 \cdot 1.3 \cdot 1.024 = \underline{1.997 \text{ HP}}$ (1)

$V = n \cdot \pi d = 1500 \text{ RPM} \cdot \frac{\pi \, 5}{12} = \underline{1963 \text{ ft/min}}$ (1)

$\therefore$ A is sufficient @ 5" sheave. (1)

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23. (6)

$F_{top} = 400\ N$

$F_{bottom} = 1100\ N$

$x_1 = 0.08\ m \quad x_2 = 0.08\ m \quad x_3 = 0.04\ cm$

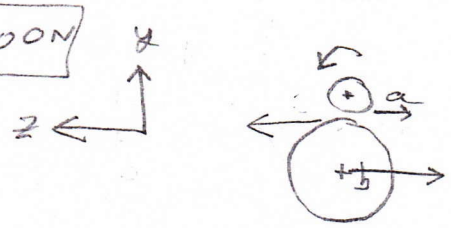
$W_T = 1200\ N$

$F_{pulley} = F_{top} + F_{bottom} = 400 + 1100 = 1500\ N$ in $+Z$ (1)

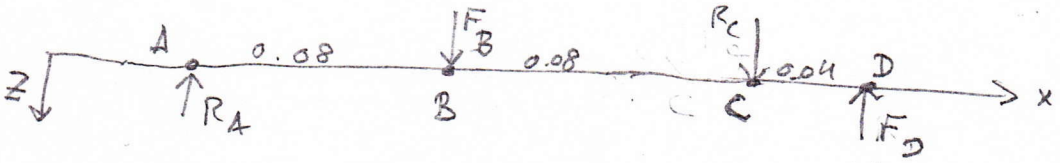
$\boxed{F_B = 1500}$

$F_B = 1200\ N$ in $-Z$ on shaft.

$\boxed{F_D = 1200\ N}$



(1)



Moment at A

$$1500 \cdot 0.08 + R_C(0.08+0.08) = 1200(0.08+0.08+0.04)$$

$$\Rightarrow R_C = [1200(0.2) - 1500(0.08)]/0.16$$

$$R_C = \frac{120}{0.16}\ N = \boxed{R_C\ 2000\ N}$$ (2)

$$R_A = F_B + R_C - F_D = 1500 + 2000 - 1200 = \boxed{R_A\ 2300\ N}$$ (2)

V2d

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24.

(8)

$x_D = 20$ $\quad a = 3$

$a_f = 1.2$ $\quad F_D = 1600\ N$

$x_0 = 0.02$ $\quad R = 0.995$

$e = 4.459$

$b = 1.483$

$$C_{10} = a_f F_D \left[\frac{x_D}{x_0 + (e - x_0)(1 - R_D)^{1/b}} \right]^{\frac{1}{a}} \quad \text{(inc. } a_f\text{)}$$

→ or: $\ln \frac{1}{R_d}$

OK if not written (2) or 1

$$= 1.2 \cdot 1600 \left[\frac{20}{0.02 + (4.459 - 0.02)(1 - 0.995)^{1/1.483}} \right]^{\frac{1}{3}}$$

$$= 1920 \left[\frac{20}{0.02 + (4.439)(0.0281)} \right]^{\frac{1}{3}} \quad (2)$$

$$= 1920 \left[\frac{20}{0.145} \right]^{\frac{1}{3}} = 1920 \cdot 5.17$$

$C_{10} = 9928\ N = 9.93\ kN$ (3)

$\therefore$ bore = 20 mm is minimum. (1)

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25.

(8)

$S_{ut} = 820$ MPa

$S_y = 650$

$S_e = 280$

$q = 0.75$ $K_T = 2.2$ $K_f = 1.75$

$q_s = 0.85$ $K_{Ts} = 3.0$ $K_{fs} = 2.2$

$n_f = 1.8$

$M_a = 95$ Nm $M_m = 0$

$T_m = 52$ Nm. $T_a = 0$

(1)

Soderberg

$$d = \left[\frac{16 n_f}{\pi} \left\{ \frac{1}{S_e} \left[4 (K_f M_a)^2 \right]^{\frac{1}{2}} + \frac{1}{S_y} \left[3 (K_{fs} T_m)^2 \right]^{\frac{1}{2}} \right\} \right]^{\frac{1}{3}}$$ (3)

$$d = \left[\frac{16 \cdot 1.8}{\pi} \left\{ \frac{1}{280\,10^6} \left[4 (1.75 \cdot 95)^2 \right]^{\frac{1}{2}} + \frac{1}{650\,10^6} \left[3 (2.2 \cdot 52)^2 \right]^{\frac{1}{2}} \right\} \right]^{\frac{1}{3}}$$ (2)

$$d = \left[9.17 \left\{ \frac{3325}{280 \cdot 10^6} + \frac{198.1}{650 \cdot 10^6} \right\} \right]^{\frac{1}{3}}$$

$$d = \left[9.17 \left\{ 1.48\,10^{-6} \right\} \right]^{\frac{1}{3}}$$

$d = 0.0236$ m $= 23.9$ mm.

(2)

Q25)

alternate solution:

$$\left(M_a = 95\ N.m \quad M_m = 0 \quad T_a = 0 \quad T_m = 52\ N.m \right) \qquad (1)$$

$$\sigma_a = K_f \frac{M_a c}{I} = 1.75\left(\frac{32(95)}{\pi d^3}\right) = \frac{1693.4}{d^3} \qquad (1)$$

$$\sigma_m = 0$$

$$\tau_a = 0$$

$$\tau_m = K_{fs} \frac{T_m c}{J} = 2.2\left(\frac{16(52)}{\pi d^3}\right) = \frac{582.6}{d^3} \qquad (1)$$

$$\sigma'_a = \sigma_a \ (1), \quad \sigma'_m = \sqrt{3}\,\tau_m \ (1)$$

$$\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_y} = \frac{1}{n} \qquad (1)$$

$$\frac{1693.4}{280 d^3} + \frac{1009.15}{650 d^3} = \frac{1}{1.8}$$

$$d^3 = 13.68$$

$$d = 2.39^{cm} \qquad (2)$$

VEL

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26. (bonus) (6)

$K_B = 576$ $\quad n = 2000$ RPM

$K_c = 0.965$ $\quad d = 5''$

$H_d = 2$ HP $\quad V = 2617$ ft/min.

$N_B = 1$ $\quad f = 0.5$

(17-21) $F_c = K_c \left(\frac{V}{1000}\right)^2 = 0.965 \left(\frac{2617}{1000}\right)^2 =$ **$F_c = 6.609$ lbf (29.4 N)** (1)

(17-22) $\Delta F = 63025 \dfrac{H_D / N_B}{n \cdot d/2} = \dfrac{2/1}{2000 \cdot 5/2} =$ **$\Delta F = 25.2$ lbf (112 N)** (2)

$\left(\Delta F = \frac{2T}{d}\right)$

(17-23) $F_1 = F_c + \Delta F \dfrac{e^{f\phi}}{e^{f\phi}-1} = 6.609 + 25.2 \dfrac{e^{0.5\pi}}{e^{0.5\pi}-1}$

$F_1 = 6.609 + 25.2 \dfrac{4.81}{3.81} = 6.609 + 31.81$ (1)

$F_1 = 38.4$ lbf. (170.9 N)

(17-24) $F_2 = F_1 - \Delta F = 38.4 - 25.2 =$ **$F_2 = 13.22$ lbf (58.8 N)** (1)

(17-25) $F_i = \dfrac{F_1 + F_2}{2} - F_c = \dfrac{38.4 + 13.2}{2} - 6.609$ (1)

$F_i = 19.20$ lbf (85.4 N)